Date : 09/09/26 Exp no : 8

Client - Server Communication using Python Socket programming

```
client.py - C:\Users\REC\client.py (3.12.10)
File Edit Format Run Options Window Help
import socket
import threading

SERVER_IP = "127.0.0.1"
PORT = 5555


def receive_messages(sock):
    while True:
        try:
            message = sock.recv(1024).decode()

            if not message:
                break

            print(f"\n{message}")
            print("You: ", end="", flush=True)

        except ConnectionResetError:
            break


client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((SERVER_IP, PORT))

print("Connected to server.")

threading.Thread(
    target=receive_messages,
    args=(client,),
    daemon=True
).start()

while True:
    message = input("You: ")

    if message.lower() == "exit":
        break

    client.sendall(message.encode())

client.close()
```

```
server.py - C:\Users\REC\server.py (3.12.10)
File Edit Format Run Options Window Help
import socket
import threading

HOST = "0.0.0.0"
PORT = 5555


def receive_messages(conn):
    while True:
        try:
            message = conn.recv(1024).decode()

            if not message:
                break

            print(f"\nClient: {message}")
            print("You: ", end="", flush=True)

        except ConnectionResetError:
            break


server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((HOST, PORT))
server.listen(1)

print(f"Server listening on port {PORT}...")

conn, addr = server.accept()
print(f"Client connected: {addr}")

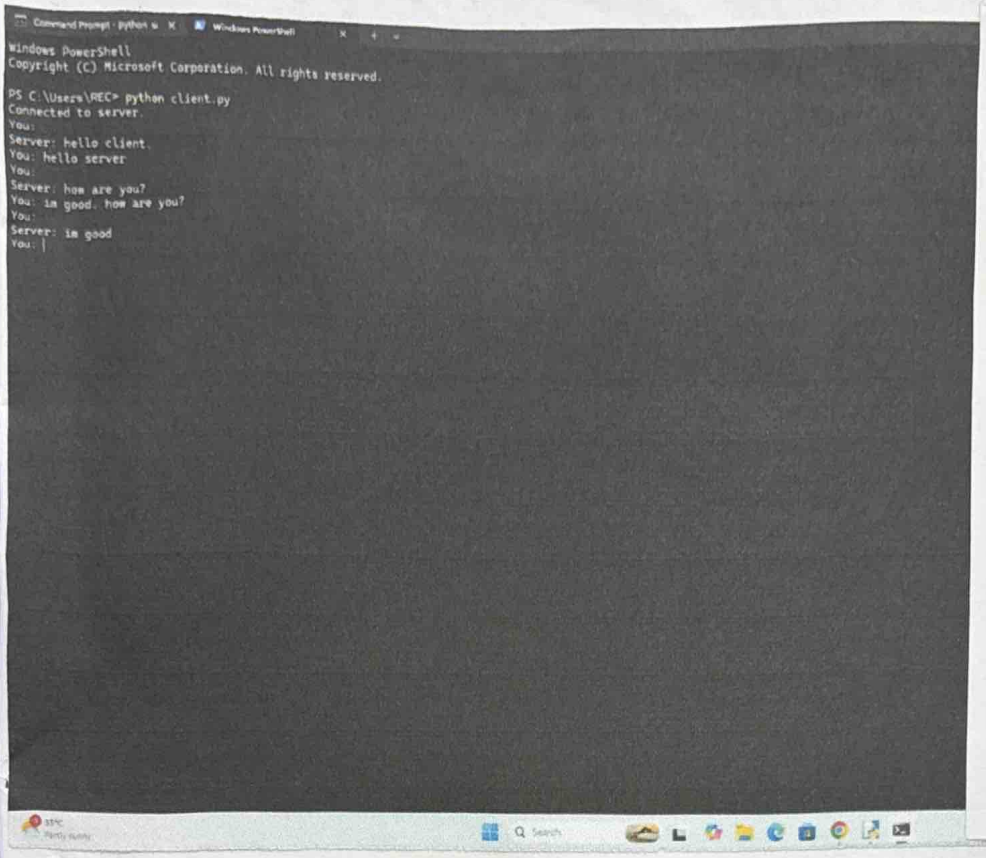
threading.Thread(
    target=receive_messages,
    args=(conn,),
    daemon=True
).start()

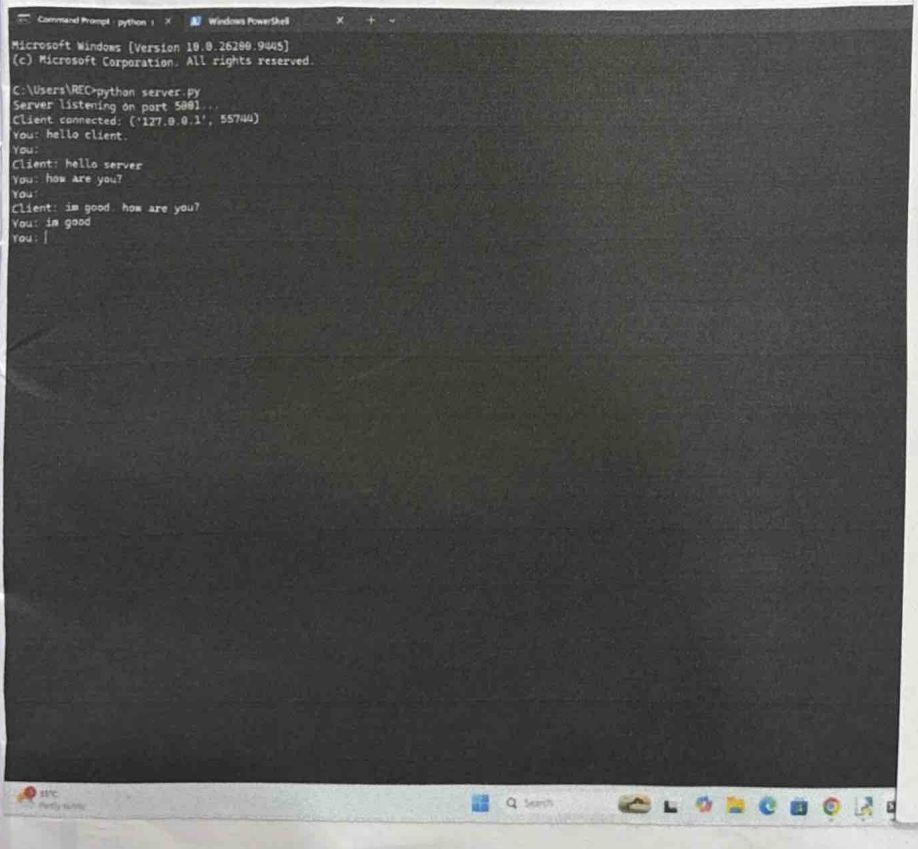
while True:
    message = input("You: ")

    if message.lower() == "exit":
        break

    conn.sendall(message.encode())

conn.close()
server.close()
```





RESULT: Thus, the client-server communication was successfully established and messages was exchanged between client and server.